



Panchip Microelectronics Co., Ltd.

PAN3029 plus PA reference design and test data

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Document Description

None

Revision History

Version	Date	Description
V1.0	2024.02	Initial version created

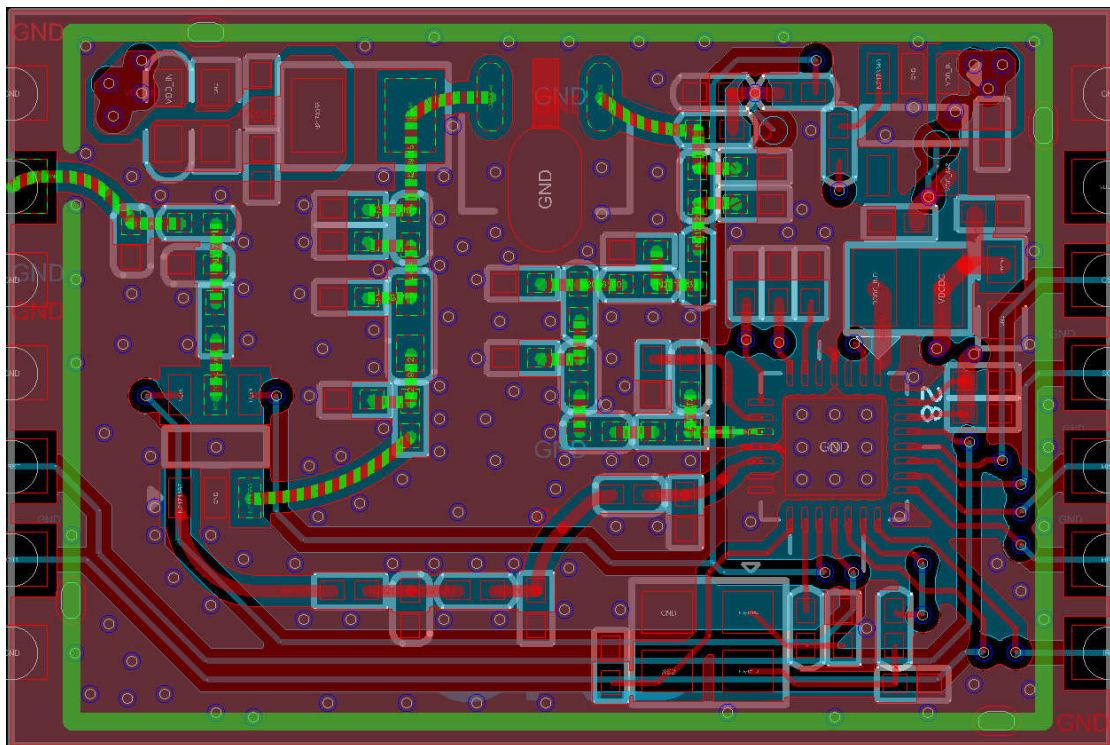
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1 PCB Reference Layout

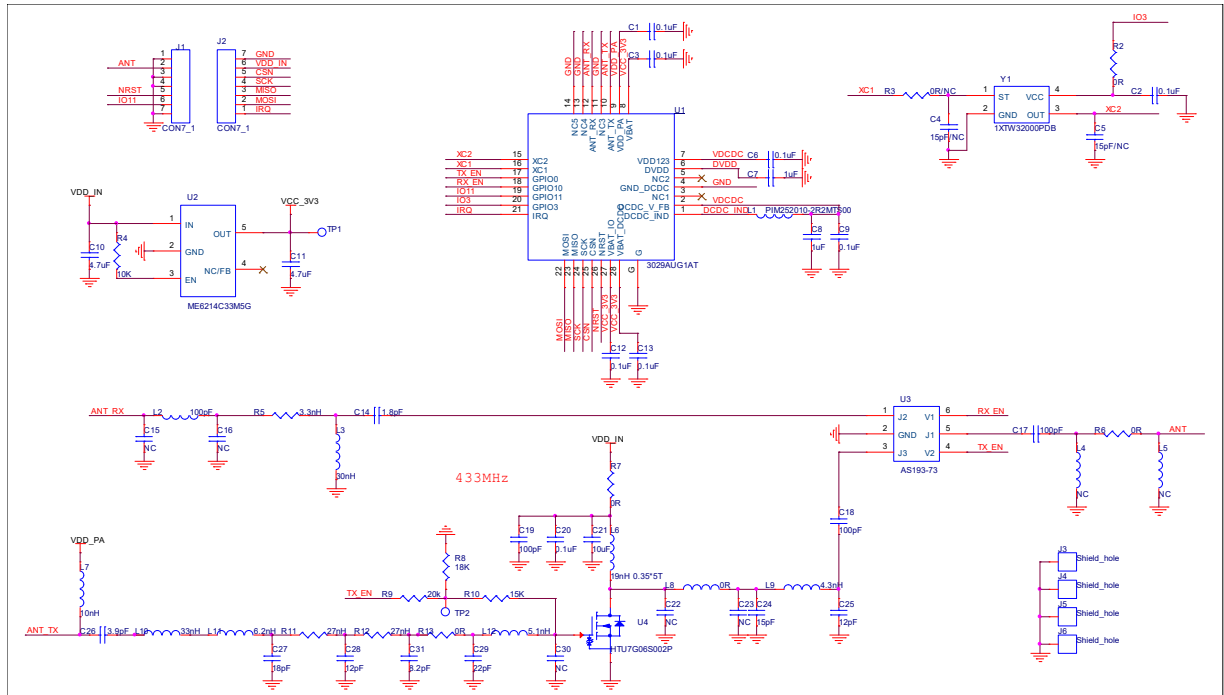
1.1 PCB reference layout for all frequency bands

The reference design for all frequency bands is debugged based on the following PCB layout. The green highlighted part is the RF transmission circuit. The VDD_IN power supply is 5V, and the PA efficiency is calculated based on the PA power consumption and the output power of the switch output.



2 433MHz Schematics and Performance

2.1 433MHz Schematic Diagram

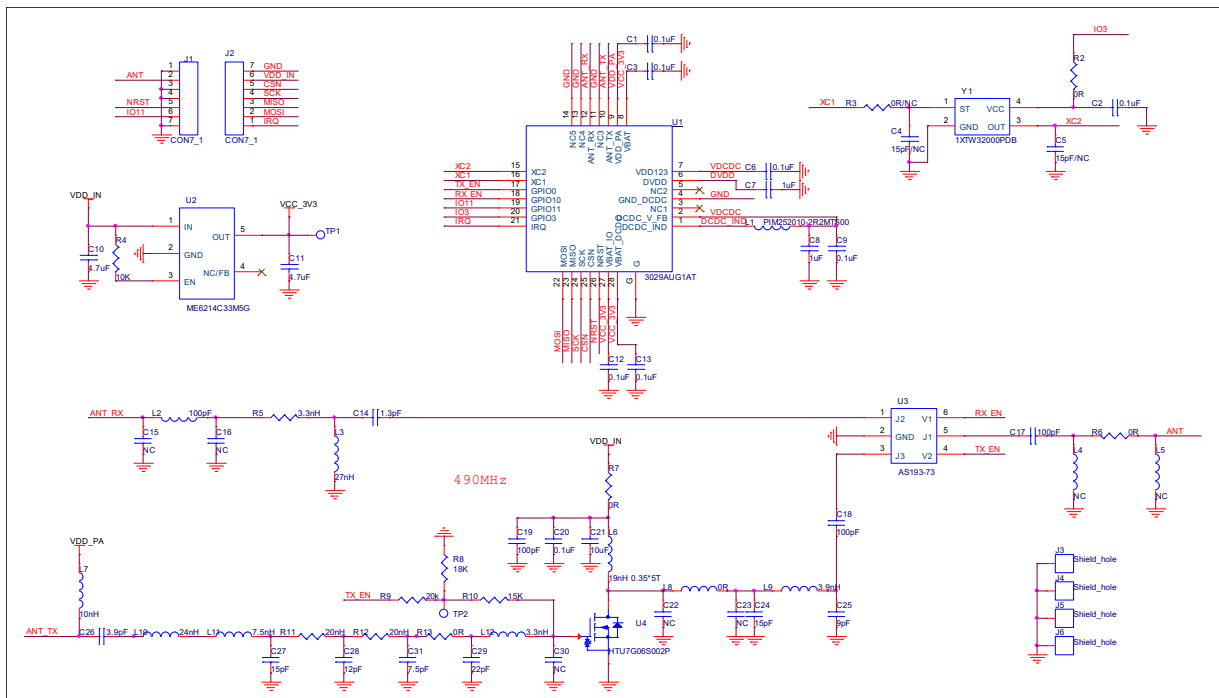


2.2 433MHz RF Performance

Frequency/MHz	Output power/dBm	Current/mA	PA efficiency
410	34.139	1125.8	55.01%
415	34.089	1080.9	55.47%
420	34.009	1050.3	56.01%
425	33.913	1029.7	56.55%
430	33.818	1015.1	56.97%
435	33.722	1005.3	57.29%
440	33.625	1000.7	57.58%
445	33.538	1002.7	57.81%
450	33.472	1010.1	57.81%
455	33.409	1013.8	57.46%
460	33.295	995.3	56.94%
465	32.769	946	54.56%
470	31.883	823	50.37%

3 490MHz Schematics and Performance

3.1 490MHz Schematic Diagram



3.2 490MHz RF Performance

Frequency/MHz	Output power/dBm	Current/mA	PA efficiency
465	33.685	1142.7	40.89%
470	33.675	1103	42.26%
475	33.634	1040	44.40%
480	33.575	981.3	46.42%
485	33.491	934.4	47.82%
490	33.408	899.2	48.75%
495	33.338	873.4	49.39%
500	33.264	854	49.66%
505	33.189	839.3	49.66%
510	33.09	827.9	49.21%
515	32.975	818.7	48.46%

935	33.503	1000.831	44.77%
940	33.509	984.559	45.57%